

Title

A Closure-Sector Spectral Model for Entanglement Correlations and Single-Particle Interference

Author

Youfu Wang (QEM framework)

ORCID: 0009-0005-5998-6024

Abstract

We propose a discrete closure-sector spectral framework (VM-24) equipped with half-cycle pairing conservation and a closure convolution operator. Under explicit and checkable assumptions, we derive: (i) no-signalling (local marginals are invariant under remote settings) as a direct consequence of pairing conservation together with a complementary local response pairing; (ii) suppression of odd harmonics and dominance of the $\cos(2\Delta)$ component as the lowest non-trivial even harmonic that survives pairing and is preferentially preserved by low-pass closure; and (iii) a shared structural parameter $A = H_2$ linking interference visibility and CHSH strength. We further show how the resulting correlation kernel can be embedded into standard Hilbert-space measurements for a singlet state, and we provide falsifiable predictions tying changes in sector-weight structure to synchronized changes in interference contrast and entanglement strength.

1. Introduction

Quantum theory predicts both single-particle interference fringes and nonlocal entanglement correlations, while maintaining no-signalling. This work provides a structural layer—based on discrete sector symmetry and closure-induced spectral filtering—explaining why (a) local marginals remain invariant under remote settings and (b) a $\cos(2\Delta)$ structure naturally emerges as the leading surviving harmonic. Our goal is not to replace quantum mechanics, but to supply an explicit structural model that (1) clarifies harmonic selection and (2) yields falsifiable cross-domain links between interference visibility and entanglement strength.

2. VM-24 Closure-Sector Model

2.1 Sector set and weights

Let the sector index set be $\Lambda = \{0, 1, \dots, 23\}$. Associate each sector λ with an angle $\theta_\lambda = \frac{\pi}{12} \lambda$.

Each experimental run induces a nonnegative weight $w_\lambda \geq 0$ with total weight $W = \sum_{\lambda \in \Lambda} w_\lambda$.

2.2 Half-cycle pairing conservation (key constraint)

The fundamental structural constraint is half-cycle pairing:

$$w_{\lambda} = w_{\lambda+12} \quad \text{for all } \lambda.$$

2.3 Spectral coefficients

Define cosine spectral coefficients (discrete cosine transform on Λ):

$$H_m = \frac{1}{W} \sum_{\lambda \in \Lambda} w_{\lambda} \cos(m\theta_{\lambda}).$$

2.4 Closure convolution as low-pass filtering (assumption)

Model closure aggregation as low-pass filtering in harmonic index m : the effective harmonic amplitude is

$$A_m = G_m H_m,$$

where $0 \leq G_m \leq 1$ and G_m is non-increasing with m (higher harmonics are attenuated at least as strongly as lower ones). This corresponds to closure-consistent averaging, which smooths high-frequency angular variations.

3. Theorem I: No-Signalling from Pairing Conservation

3.1 Local response and marginals

Let $A(a, \lambda) \in \{+1, -1\}$ denote Alice's deterministic outcome rule given local setting a and sector λ , and $B(b, \lambda)$ similarly for Bob. Alice's local marginal is

$$P(A=+1 \mid a) = \frac{1}{W} \sum_{\lambda} w_{\lambda} \mathbf{1}[A(a, \lambda)=+1].$$

3.2 Complementary local pairing (explicit)

Assume the half-cycle complement flips the local sign:

$$A(a, \lambda+12) = -A(a, \lambda), \quad B(b, \lambda+12) = -B(b, \lambda).$$

This is the discrete analogue of a π -shift mapping to a complementary binary readout.

3.3 Theorem and proof

Theorem 1 (No-signalling). If (i) $w_{\lambda} = w_{\lambda+12}$ and (ii) $A(a, \lambda+12) = -A(a, \lambda)$ (and similarly for B), then $P(A=+1 \mid a) = 1/2$ for all a , and analogously for Bob. Hence local marginals are invariant under remote settings (no-signalling).

Proof. Pair λ with $\lambda+12$. Since the outcomes are complementary, exactly one element in each pair yields $+1$. Therefore

$$\mathbf{1}[A(a, \lambda)=+1] + \mathbf{1}[A(a, \lambda+12)=+1] = 1.$$

Using $w_{\lambda} = w_{\lambda+12}$, the weighted sum over each pair contributes w_{λ} . Summing over all 12 pairs gives $\sum_{\lambda} w_{\lambda} \mathbf{1}[A(a, \lambda)=+1] = W/2$. Dividing by W yields $P(A=+1 \mid a) = 1/2$. $\square$

4. Theorem II: Harmonic Suppression and $\cos(2\Delta)$ Dominance

4.1 Odd-harmonic cancellation

Because $\theta_{\lambda+12} = \theta_{\lambda} + \pi$,
 $\cos(m(\theta_{\lambda} + \pi)) = (-1)^m \cos(m\theta_{\lambda})$.

Pairing implies odd m terms cancel exactly, hence

$$H_m = 0 \quad \text{for all odd } m.$$

4.2 Lowest surviving even harmonic and low-pass closure

Since all odd m vanish, the lowest non-trivial harmonic is $m=2$. With low-pass closure G_m non-increasing in m , higher even harmonics $m \geq 4$ are attenuated at least as strongly. Under mild regularity (sector weights not alternating rapidly), $m=2$ becomes dominant.

4.3 Correlation kernel

Let Δ denote the effective phase difference between two settings (or paths). The closure-sector kernel has the harmonic form

$$K(\Delta) = \sum_{m \geq 0} A_m \cos(m\Delta), \quad A_m = G_m H_m.$$

Thus to leading order,

$$K(\Delta) \approx A \cos(2\Delta) + B \cos(4\Delta) + \dots, \quad A = A_2.$$

5. Single-Particle Interference Mapping

5.1 Geometric mapping $x \mapsto \Delta(x)$

For a standard double-slit geometry under far-field approximation,

$$\Delta(x) = \frac{\pi d}{\lambda L} x,$$

where d is slit separation, λ wavelength, and L distance to the screen.

5.2 Intensity and visibility

Assuming a baseline single-particle visibility parameter V (instrumental/coherence), the model predicts

$$I(x) \propto 1 + V |A| \cos(2\Delta(x)).$$

The experimental fringe visibility $V_{\text{exp}} = (I_{\text{max}} - I_{\text{min}}) / (I_{\text{max}} + I_{\text{min}})$ satisfies

$$V_{\text{exp}} = V |A|.$$

6. Hilbert-Space Embedding and CHSH Strength

6.1 Pauli embedding

Let σ_x, σ_z be Pauli matrices and define

$$\sigma(\theta) = \cos(2\theta) \sigma_z + \sin(2\theta) \sigma_x.$$

For the singlet state $|\psi\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$,
 $E(a,b) = \langle \psi | \sigma(a) \otimes \sigma(b) | \psi \rangle = -\cos(2(a-b))$.
 To incorporate sector amplitude, set
 $E_{\text{A}}(a,b) = -A \cos(2(a-b))$, $|A| \leq 1$.

6.2 CHSH (clean derivation)

Use the CHSH form

$$S = \big| E(a,b) + E(a,b') + E(a',b) - E(a',b') \big|.$$

Choose $a=0$, $a'=\pi/4$, $b=\pi/8$, $b'=-\pi/8$. Then

$$E(a,b) = -A \cos(\pi/4) = -\frac{A}{\sqrt{2}}, \quad$$

$$E(a,b') = -\frac{A}{\sqrt{2}}, \quad$$

$$E(a',b) = -\frac{A}{\sqrt{2}}, \quad$$

$$E(a',b') = -A \cos(3\pi/4) = +\frac{A}{\sqrt{2}}.$$

Therefore

$$S = \left| -\frac{A}{\sqrt{2}} - \frac{A}{\sqrt{2}} - \frac{A}{\sqrt{2}} + \frac{A}{\sqrt{2}} \right| = 2\sqrt{2}|A|.$$

Hence the Tsirelson bound is achieved when $|A|=1$.

6.3 Unified falsifiable linkage

Combining interference and CHSH yields

$$\frac{S}{2\sqrt{2}} = |A| = \frac{V_{\text{exp}}}{V}.$$

Thus controlled changes in sector-weight structure w_{λ} that modify $|A|$ should co-vary interference visibility and entanglement strength.

7. Conclusion

We introduced a closure-sector spectral model on VM-24 with half-cycle pairing conservation and low-pass closure. Under explicit assumptions, we proved no-signalling, showed odd-harmonic cancellation, and justified $\cos(2\Delta)$ dominance as the leading surviving harmonic. We embedded the kernel into standard Hilbert-space measurements for a singlet and derived an explicit CHSH expression $S = 2\sqrt{2}|A|$. The model yields falsifiable predictions linking interference visibility and entanglement strength via the shared structural parameter $A = H_2$.

Appendix A. Minimal assumptions checklist

A1. Sector pairing: $w_{\lambda} = w_{\lambda+12}$.

A2. Local response complement: $A(a, \lambda+12) = -A(a, \lambda)$,
 $B(b, \lambda+12) = -B(b, \lambda)$.

A3. Closure low-pass: $A_m = G_m H_m$ with G_m non-increasing in m .

A4. (Sufficient for strong dominance) Sector weights are smooth (no alternating high-frequency

pattern).

References (add before submission)

- [1] J. S. Bell, “On the Einstein Podolsky Rosen paradox,” *Physics* 1, 195–200 (1964).
- [2] J. F. Clauser, M. A. Horne, A. Shimony, R. A. Holt, “Proposed experiment to test local hidden-variable theories,” *Phys. Rev. Lett.* 23, 880 (1969).
- [3] B. S. Tsirelson, “Quantum generalizations of Bell’s inequality,” *Lett. Math. Phys.* 4, 93–100 (1980).
- [4] Standard optics/quantum texts for double-slit far-field mapping (choose one appropriate reference).